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Monday

Assignment-03

Set 1

CSA5117

cryptography &
Network Security

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Consider the following two hash algorithm: MD5 and SHA-1.

Both are used to produce message digests.

a) compare the output lengths of MD5 and SHA-1.

- MD5 (Message Digest Algorithm 5):

produces a 128-bit hash value, which is usually represented as 32 hexadecimal characters.

- SHA-1 (Secure Hash algorithm 1):

produces a 160-bit hash value, represented as 40 hexadecimal characters.

Algorithm	output length	Hexadecimal Representation
MD5	128 bits	32 characters
SHA1	160 bits	40 characters

Therefore, SHA-1 produces a longer hash than MD5.

b) Explain why MD5 is no longer considered secure.

MD5 is no longer considered secure because it has serious collision vulnerabilities. A collision occurs when two different messages produce the same hash value.

Researchers have demonstrated practical methods for deliberately creating two different files or messages with the same MD5 hash. This makes MD5 unsuitable for security applications such as digital signatures, certificates, and verifying the authenticity of important files.

Although MD5 can still be used for non-security purposes such as detecting accidental data corruption, it should not be used where strong security is required.

In MD5, researchers have demonstrated practical techniques for creating different messages with the same MD5 hash.

This is a serious security problem because an attacker may create a malicious file that has the same hash as a legitimate file.

For example:

legitimate file



MD5



ABC123456789



Malicious file



MD5



ABC123456789

Although The two files are different, they have the same MD5 digest.

This weakness can be dangerous when MD5 is used for:

- Digital signatures
- Digital certificate
- Software integrity verification
- Authentication systems
- Security sensitive file verification.

An attacker could potentially substitute a malicious document for a legitimate document while keeping the same MD5 hash.

MD5 is therefore not recommended for security-related applications. It may still be useful for detecting accidental changes in data, but stronger

algorithms such as SHA-256 or SHA-3 should be used for cryptographic security.

C. If two different files result in the same hash in MD5, what type of attack does this represent?

If two different files produce the same MD5 hash value, it represents a collision attack.

A collision occurs when:

$$M_1 \neq M_2 \text{ but}$$

$$\text{MD5}(M_1) = \text{MD5}(M_2)$$

where M_1 and M_2 are two different messages or files.

for example:

File A = "Important document"

↓

MD5

↓

5D41402ABC4B2A76B97719D91

File B = "Malicious document"

↓

MD5

↓

5D41402ABC4B2A76B97719D91

Here File A and File B are different, but their hash values are the same. This is called a hash collision.

A collision attack specifically attempts to find any two different inputs with the same hash value. It is different from a preimage attack, where the attacker tries to find an input corresponding to a particular hash value.

Since practical collision attacks against MD5 have been demonstrated, MD5 cannot be considered reliable for applications that depend on collision resistance.

d) Why is SHA-256 considered more secure than SHA-1?

Illustrate with reference to collision resistance.

SHA-256 is considered more secure than SHA-1 because it produces a 256-bit hash value, while SHA-1 produces a 160-bit hash value. More importantly, SHA-1 has suffered practical collision attacks, whereas no practical collision attack against SHA-256 is currently known.

For an ideal hash function with an n -bit output, the birthday attack suggests that finding a collision requires approximately:

$$2^{n/2} \text{ operations.}$$

For SHA-1:

$$n=160$$

Therefore, the theoretical generic collision resistance is approximately:

$$2^{80}$$

For SHA-256:

$$n=256$$

Therefore, the theoretical generic collision resistance is approximately:

$$2^{128}$$

Thus, SHA-256 provides a significantly larger security margin against generic collision attacks.